Argon System Conceptual Review

Find out where your purchased argon actually goes. Map it, measure it, tag the leaks, price the fixes.

What it is

Inflow measures how much argon the plant buys, how much reaches the process, and how much leaks, vents, boils off, or flows while nothing is running. Same method Inflow has used on compressed air for 18 years, applied to a gas you pay for by the cubic foot.

What the review covers

- **System map.** Walk down and map the system from the bulk tank tie-in through regulators to every point of use. Block diagram with regulator setpoints and pipe sizes.
- **Cost baseline.** 12 month usage and cost baseline from delivery invoices and tank telemetry at your contract gas rate.
- **Idle draw.** Tank level readings across a shutdown (overnight or weekend) separate leaks, purge flow and boil off from productive use.
- **Leak survey.** Ultrasonic survey. Each leak is tagged, sized, and valued at your delivered argon price.
- **Point of use and regulator survey.** Shielding flow versus welder specification, pre and post flow times, open or abandoned drops, uncoupled quick disconnect whips, rubber hose runs, dead legs, regulator creep.
- **Moisture path review.** If points of use fail purity tests, trace where moisture enters between the tank and the failing point: hose materials, dead legs, idle lines, purge practice, and the test method itself.
- **Report.** System map, annual baseline, tagged leak list with $/yr, and a ranked table of fixes with savings and budget costs.

Who it is for

Laser and TIG welding cells, insulated glass argon fill lines, gloveboxes and purge stations, ICP and lab argon, and any plant on a micro bulk or bulk argon tank that has never read the tank level over a weekend.

Proof point

A Minnesota metal fabrication plant asked for the review because nobody could say where the argon went. Two site visits and one overnight tank reading later:

Item	Result
Argon purchased	About 300,000 CF per year, $33,000 per year
Draw with the plant shut down	About two thirds of the average draw
Cause	Several small leaks, open quick disconnect whips, tank boil off. No single large leak.
Recoverable	About $15,000 per year
Review fee and payback	$4,700, about 4 months

What you provide

Escorted access to argon lines. 12 months of delivery invoices and the contract rate. Piping drawings and an equipment list if any. Supplier certificate of analysis and telemetry access. A shutdown window for the tank reading.

Fee

Scope	Fee
Argon System Conceptual Review, two site visits, leak survey, report	$4,700
Additional building or system on the same campus, same trip	$1,500 each
Optional temporary flow metering on the main line (rental)	quote

All inclusive, no travel or hourly charges. Delivered argon carries no utility rebate, so the fee pays back out of the gas bill.

How it works

- **1. Observe.** Walk the system in production. Decide what needs measuring and where the tank reading will come from.
- **2. Measure.** Invoices, telemetry, an idle period tank reading, and an ultrasonic pass on every accessible line and fitting.
- **3. Report.** Findings, tagged leaks, ranked fixes with dollars and budget costs, in a document your team and your gas supplier can both use.

Typical schedule: two site days plus one idle period tank reading. Draft report within two weeks of the second visit.

Where argon goes at idle

- Welder pre and post flow while the cell sits between parts
- Quick disconnect whips left uncoupled at the machine
- Regulator outlet fittings and old copper drops
- Micro bulk tank boil off venting to the line or relief
- Purge stations left on between shifts

None of these show up on a bill. The tank level does.

Credentials
Xcel Energy Energy Efficiency Partner Award winner, 12 consecutive years, 2015 to 2026
U.S. Department of Energy qualified AirMaster+, Pump and Fan System Specialist
Independent. Vendor neutral. We measure it, we do not sell it. 20+ years on industrial systems
Xcel Energy listed study provider, Minnesota compressed air and process efficiency programs

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